

Symbiotic model - Updated 11.12.25

Astrangia poculata sym

COMPLETE = 3.0; MRE = 0.076; SMAE = 0.108; SMSE = 0.040

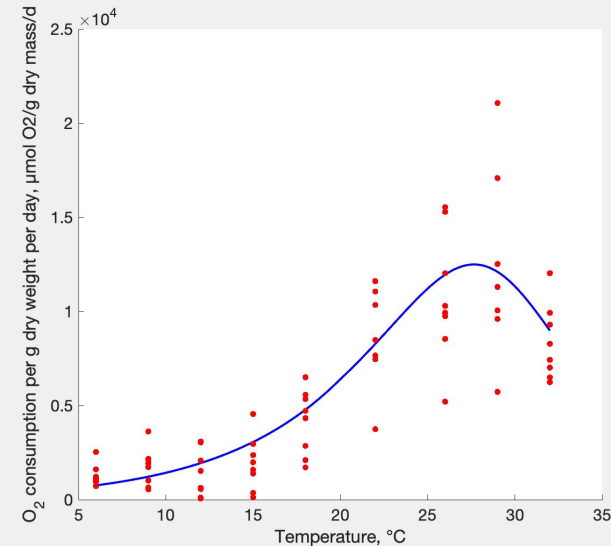
Data & predictions

data	prd	RE	symbol	units	description
1096	1096	0.0001763	tp	d	time since birth at puberty
0.011	0.01101	0.0008916	L0	cm	egg diameter
0.0137	0.01373	0.001877	Lb	cm	total length at birth
0.8	0.7619	0.04762	RQ	-	Respiratory quotient
see figure		0.3058	TJO	Temperature	vs. O ₂ consumption per g dry weight per day
0.02	0.02665	0.3325	psd.v	cm/d	energy conductance
18	80	3.444	psd.p_M	J/d.cm ³	vol-spec som maint
0.002	0.002	0	psd.k_J	1/d	maturity maint rate coefficient
0.3	0.006521	0.9783	psd.k	-	maintenance ratio
0.8	0.6922	0.1347	psd.kap	-	allocation fraction to soma
0.8	0.802	0.002503	psd.kap_G	-	growth efficiency
0.95	0.95	0	psd.kap_R	-	reproduction efficiency



std parameters at 20.0°C

symbol	units	value	free	description
T_ref	K	293.1	0	Reference temperature
T_A	K	1.246e+04	1	Arrhenius temperature
z	-	0.0727	1	zoom factor
F_m	l/d.cm ²	6.5	0	{F_m}, max spec searching rate
kap_X	-	0.8	0	digestion efficiency of food to reserve
kap_P	-	0.1	0	faecation efficiency of food to faeces
v	cm/d	0.02665	1	energy conductance
kap	-	0.6922	1	allocation fraction to soma
kap_R	-	0.95	0	reproduction efficiency
p_M	J/d.cm ³	80	0	[p_M], vol-spec somatic maint
p_T	J/d.cm ²	0	0	{p_T}, surf-spec somatic maint
k_J	1/d	0.002	0	maturity maintenance rate coefficient
E_G	J/cm ³	260.9	1	[E_G], spec cost for structure
E_Hb	J	9.644e-11	1	maturity at birth
E_Hp	J	3.4	1	maturity at puberty
h_a	1/d ²	1.282e-15	1	Weibull aging acceleration
s_G	-	0.0001	0	Gompertz stress coefficient
T_AH	K	3.501e+04	1	Arrhenius temperature for upper boundary
T_H	K	302.4	0	upper boundary tolerance range
del_M	-	0.006848	1	shape coefficient for larvae
del_W	-	1	0	dry - ash free dry weight ratio
del_Y	-	0.01233	1	shape coefficient for egg
f	-	1	0	scaled functional response for 0-var data



Aposymbiotic model - Updated 11.12.25

Astrangia poculata apo

COMPLETE = 3.0; MRE = 0.097; SMAE = 0.147; SMSE = 0.041

Data & predictions

data	prd	RE	symbol	units	description
1096	1057	0.03561	tp	d	time since birth at puberty
0.011	0.01069	0.02847	L0	cm	egg diameter
0.0137	0.01398	0.02039	Lb	cm	total length at birth
0.8	0.7619	0.04762	RQ	-	Respiratory quotient
see figure		0.3153	TJO	Temperature	vs. O ₂ consumption per g dry weight per day
0.02	0.05092	1.546	psd.v	cm/d	energy conductance
18	148.7	7.259	psd.p_M	J/d.cm ³	vol-spec som maint
0.002	0.002	0	psd.k_J	1/d	maturity maint rate coefficient
0.3	0.003407	0.9886	psd.k	-	maintenance ratio
0.8	0.5499	0.3126	psd.kap	-	allocation fraction to soma
0.8	0.8262	0.03269	psd.kap_G	-	growth efficiency
0.95	0.95	0	psd.kap_R	-	reproduction efficiency

std parameters at 20.0°C

symbol	units	value	free	description
T_ref	K	293.1	0	Reference temperature
T_A	K	1.654e+04	1	Arrhenius temperature
z	-	0.03505	1	zoom factor
F_m	l/d.cm ²	6.5	0	{F_m}, max spec searching rate
kap_X	-	0.8	0	digestion efficiency of food to reserve
kap_P	-	0.1	0	faecation efficiency of food to faeces
v	cm/d	0.05092	1	energy conductance
kap	-	0.5499	1	allocation fraction to soma
kap_R	-	0.95	0	reproduction efficiency
p_M	J/d.cm ³	148.7	1	[p_M], vol-spec somatic maint
p_T	J/d.cm ²	0	0	{p_T}, surf-spec somatic maint
k_J	1/d	0.002	0	maturity maintenance rate coefficient
E_G	J/cm ³	253.2	1	[E_G], spec cost for structure
E_Hb	J	2.345e-10	1	maturity at birth
E_Hp	J	1.749	1	maturity at puberty
h_a	1/d ²	1.073e-15	1	Weibull aging acceleration
s_G	-	0.0001	0	Gompertz stress coefficient
T_AH	K	4.059e+04	1	Arrhenius temperature for upper boundary
T_H	K	300.4	0	upper boundary tolerance range
del_M	-	0.00745	1	shape coefficient for larvae
del_W	-	1	0	dry - ash free dry weight ratio
del_Y	-	0.01376	1	shape coefficient for egg
f	-	1	0	scaled functional response for 0-var data

